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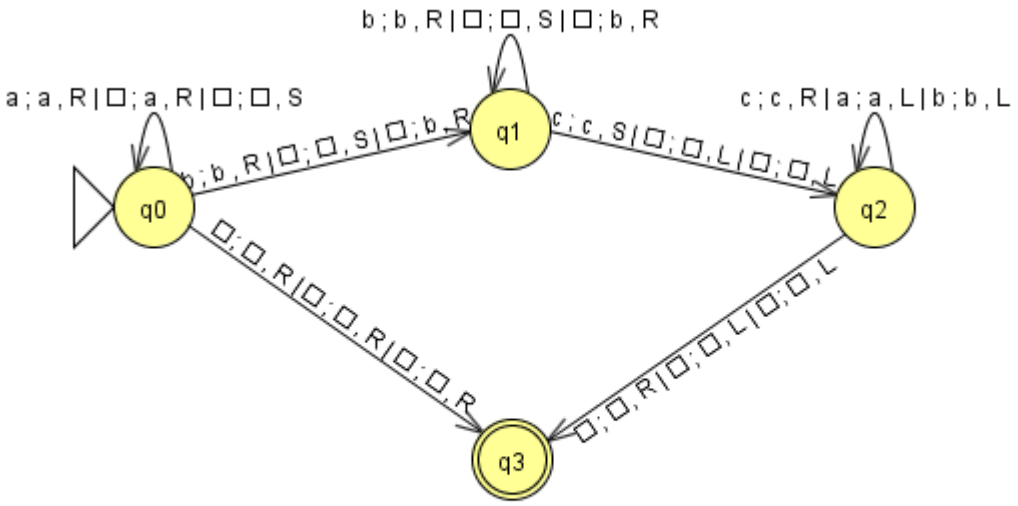
Universidade de Vila Velha
14 de abril de 2026

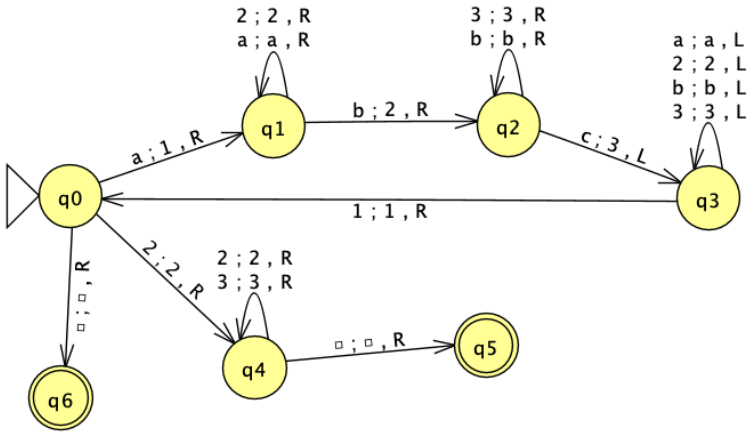
NO MEU COMPUTADOR A VERSÃO 8 ESTAVA BUGADA E NÃO ESTAVA APARECENDO A OPÇÃO DE STOP MESMO QUANDO MUDAVA NAS CONFIGURAÇÕES, ENTÃO FIZEMOS NA VERSÃO 7 QUE RODAVA PARA NÓS DUAS.

Trabalho sobre Máquina de Turing

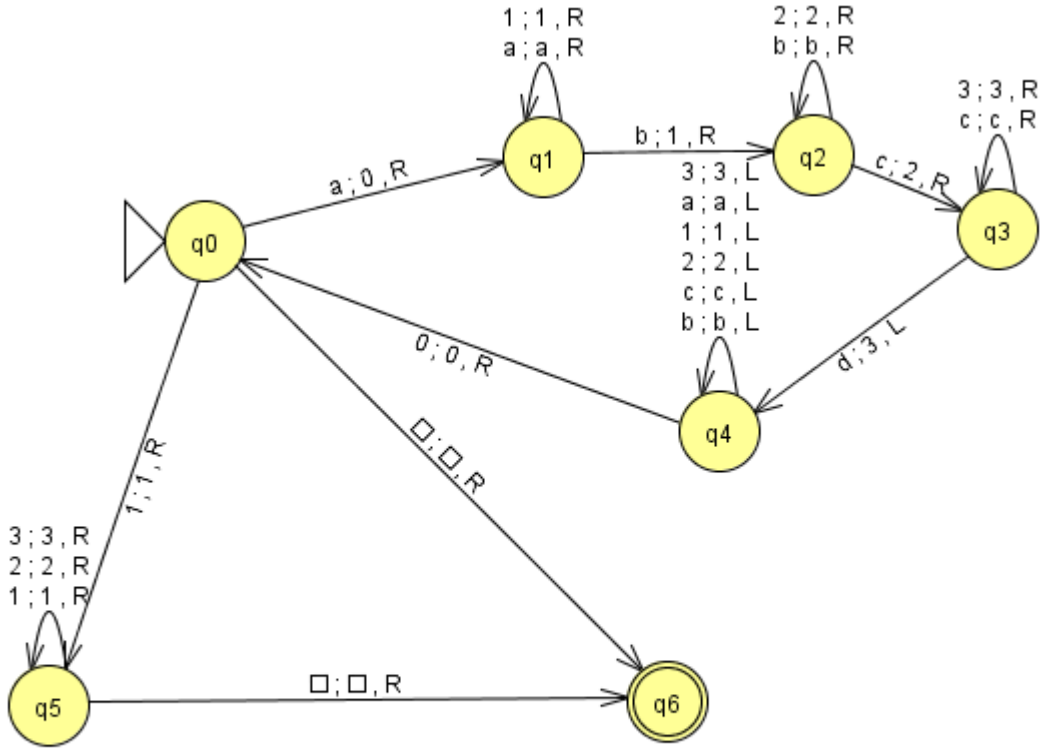
1) fazer a máquina de touring com 3 fitas para reconhecer L={anbncn , n>=0} Σ = {a,b,c}



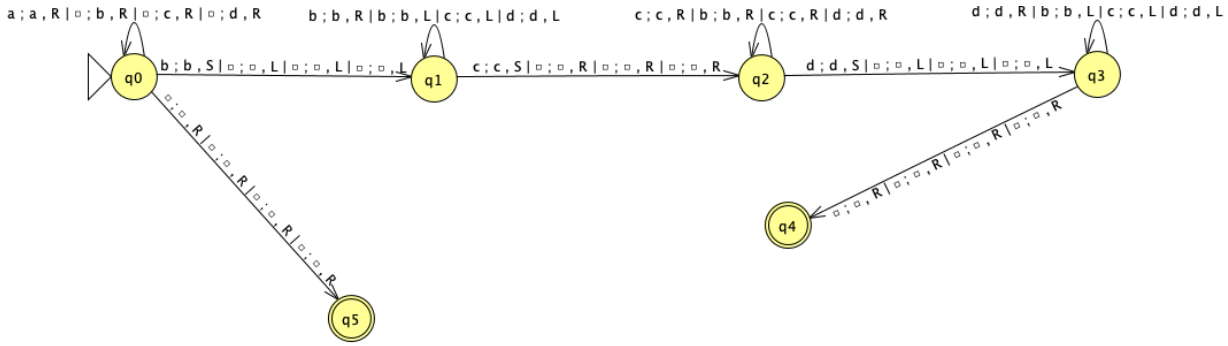
2) fazer a máquina de touring com 1 fita para reconhecer L={anbncn , n>=0} Σ = {a,b,c}



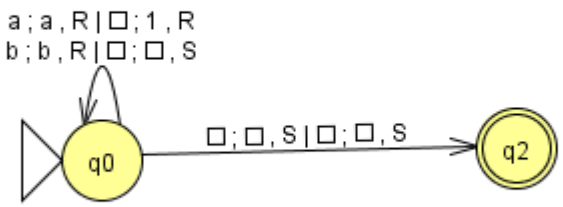
3) fazer a máquina de touring com 1 fita para reconhecer L={ anbncndn , n>=0} Σ = {a,b,c,d}



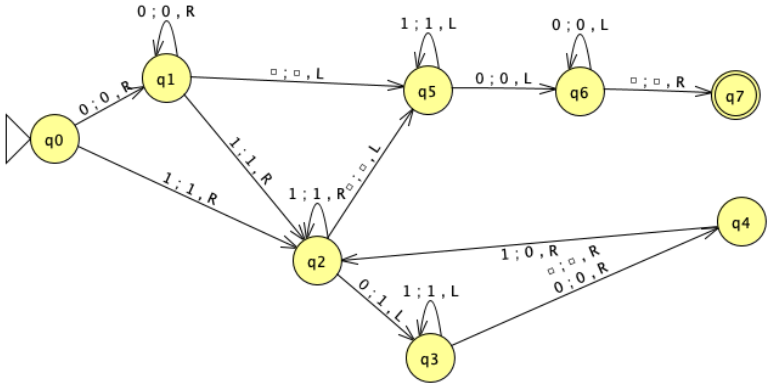
4) fazer a máquina de touring com 4 fitas para reconhecer L={ anbncndn , n>=0} Σ = {a,b,c,d}



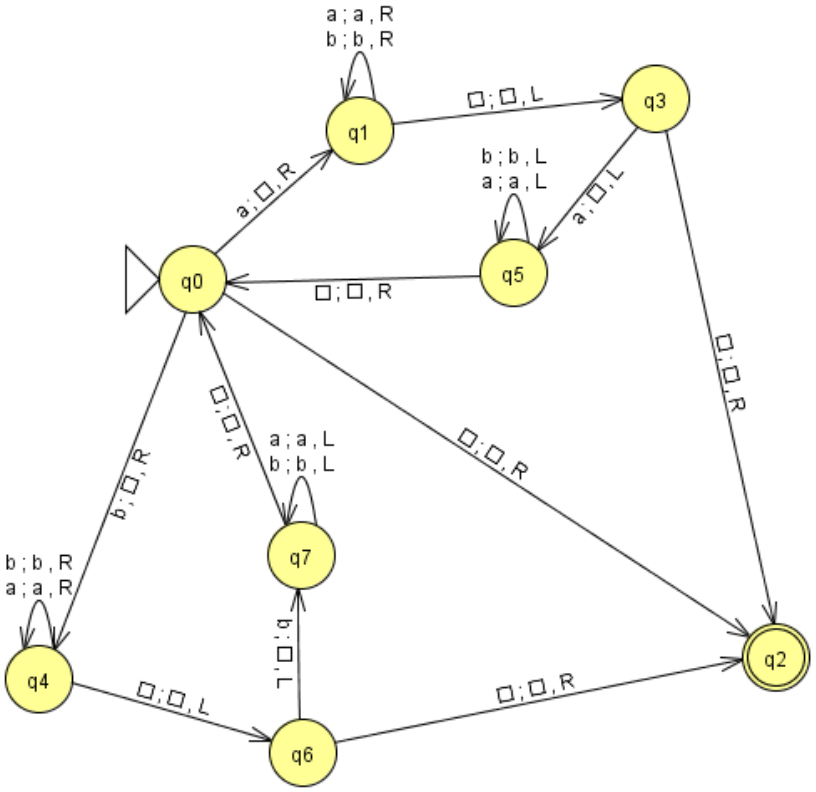
5) Construa uma máquina de Turing que conte o número de a's de uma cadeia de caracteres (Σ = {a, b}). A MT é iniciada com a cabeça sobre o primeiro carácter da cadeia (ex. q0 abababbba). Deve parar à direita da cadeia com a cabeça a apontar para o número de ocorrências do carácter a (abababbba□ qf 1111).



6) Construa uma MT que dada uma cadeia de caracteres (Σ = {0, 1}) ordene essa cadeia. A MT é iniciada com a cabeça sobre o primeiro carácter da cadeia (ex. q0 1000110) e deve parar com a cabeça sobre o primeiro carácter da cadeia ordenada(ex. qf 0000111).

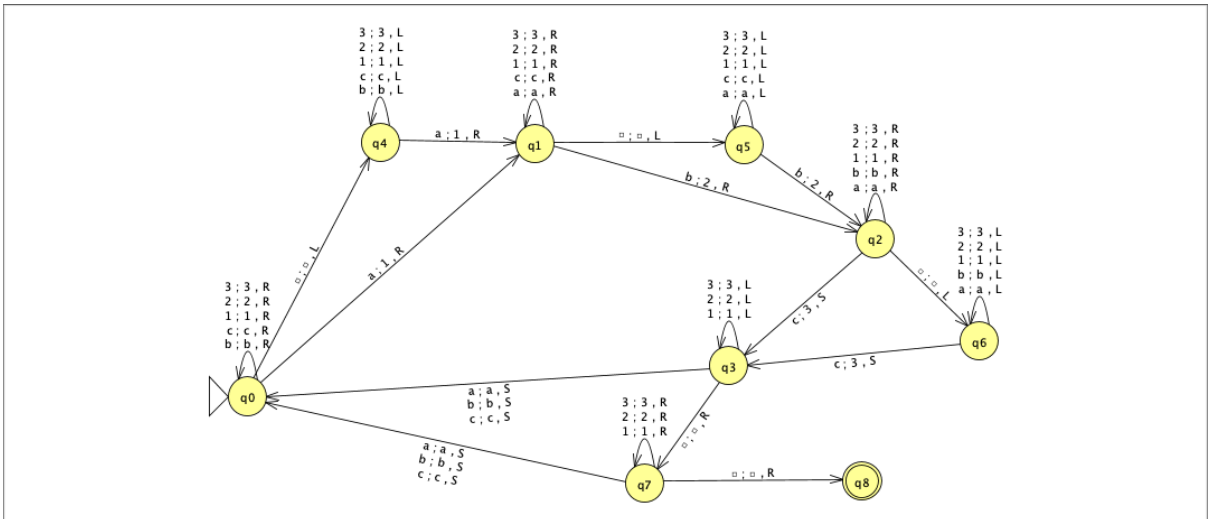


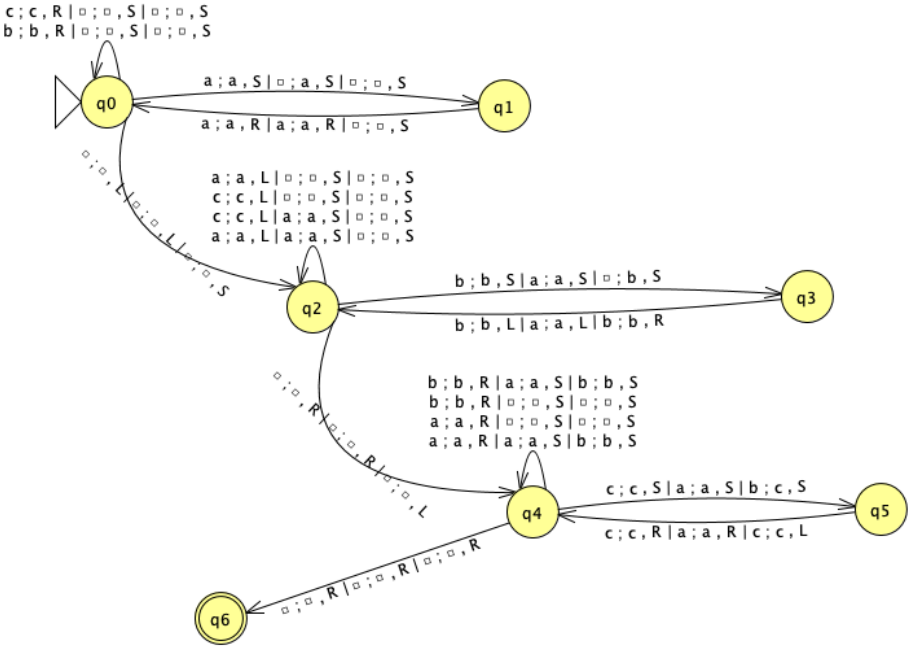
7) Construa uma MT (Σ = { a, b}) que detecte se uma cadeia é um palíndromo.



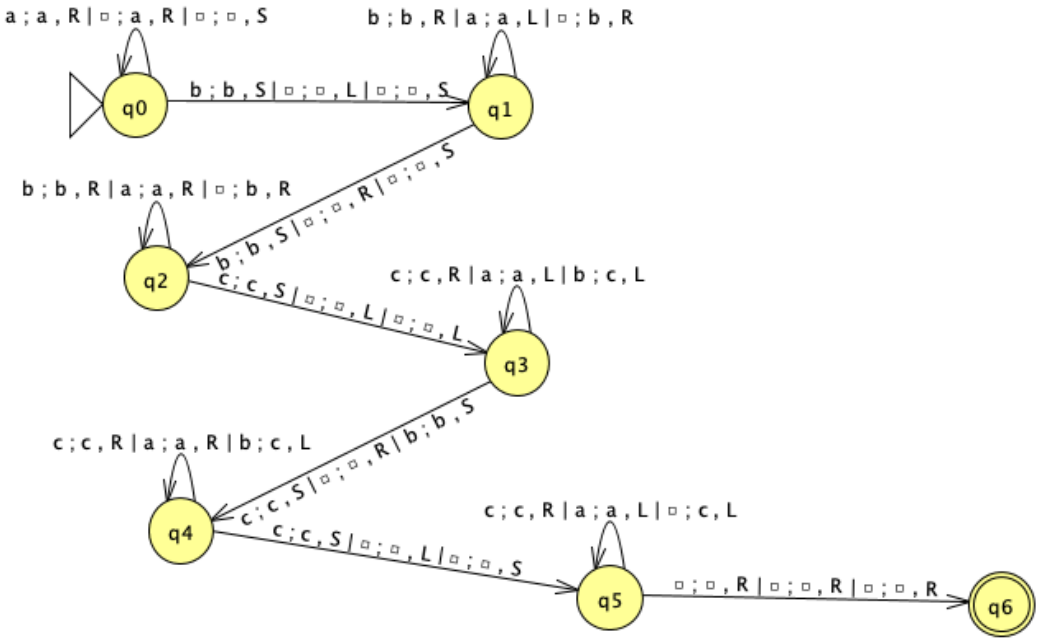
8) Construa uma MT que reconheça as seguintes linguagens:

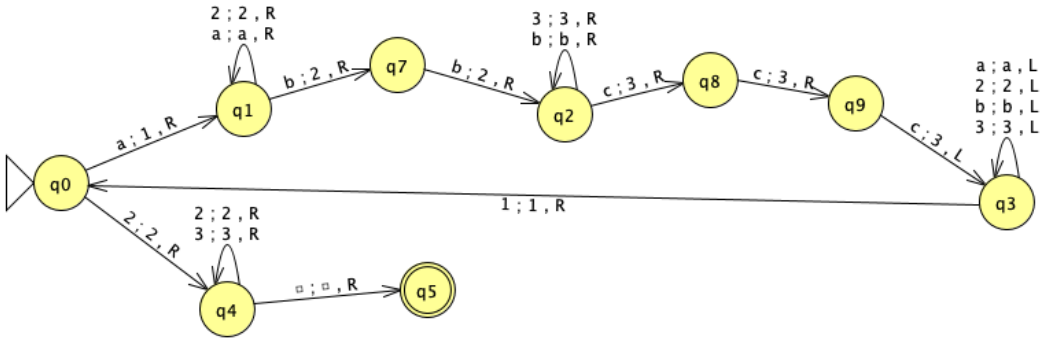
a) $L = \{ w \mid na(w) = nb(w) = nc(w)\}$ fazer com 1 fita e com 3 fitas. Σ = {a,b,c}



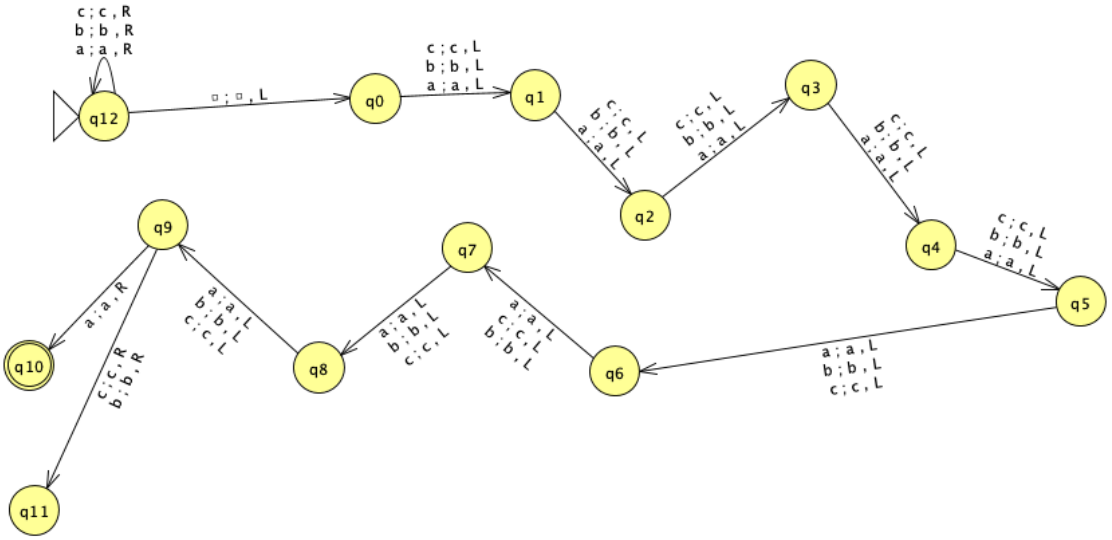


b) $L = \{ a^n b^{2n} c^{3n} \mid n \geq 0 \}$ fazer com 1 fita e com 3 fitas. Σ = {a,b,c}

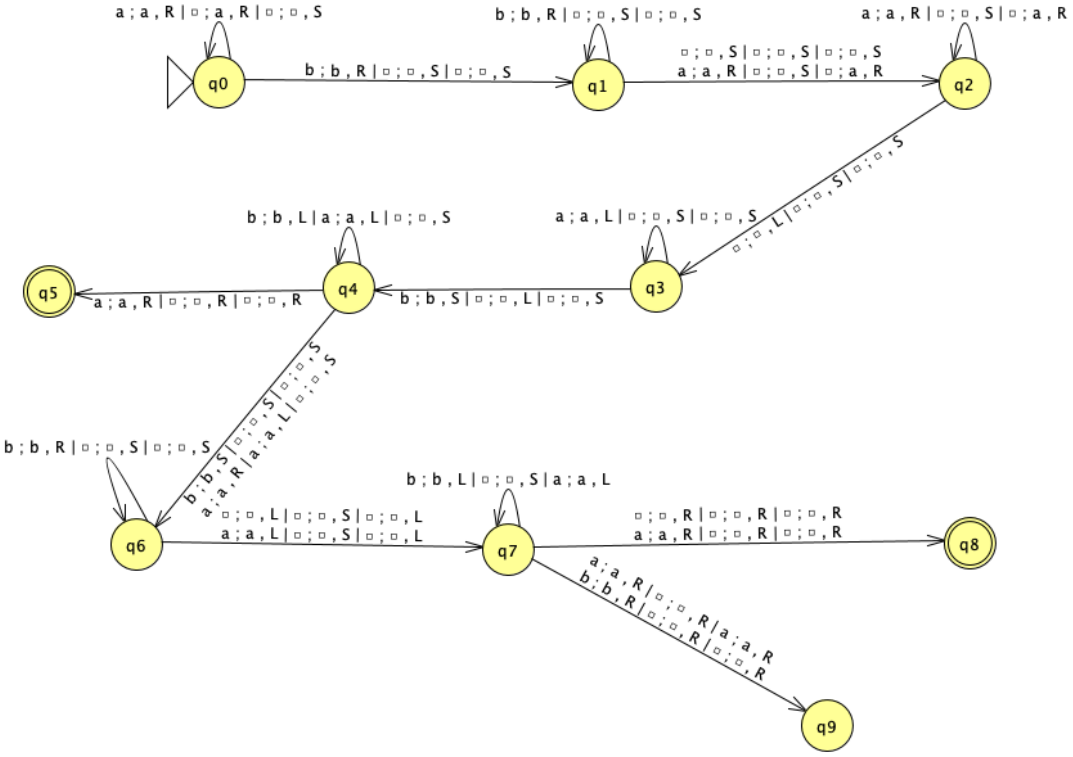


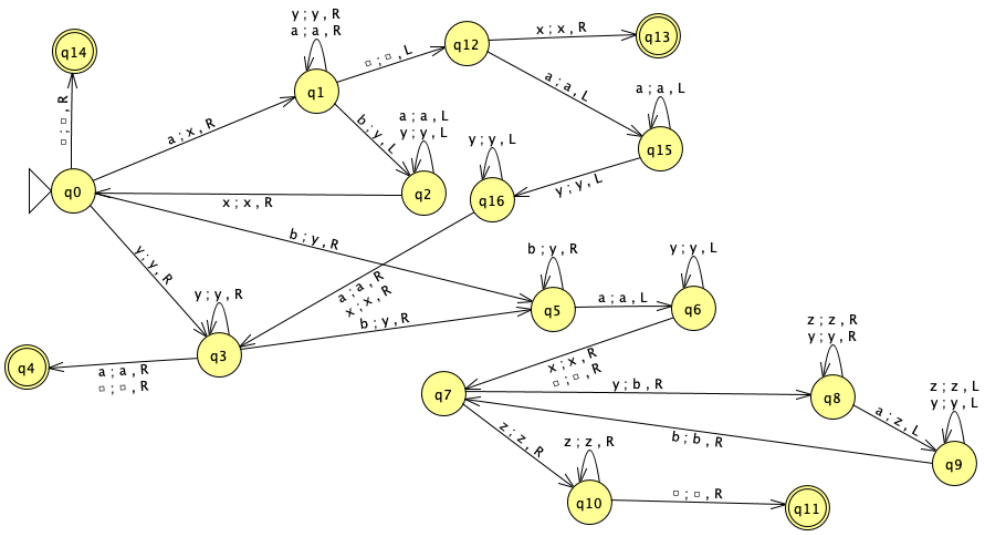


c) $L_4 = \{ w \mid$ o décimo símbolo da direita para a esquerda é a $\}$ fazer com 1 fita Σ = {a,b,c}

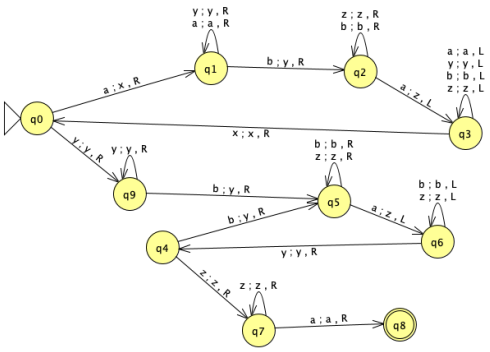


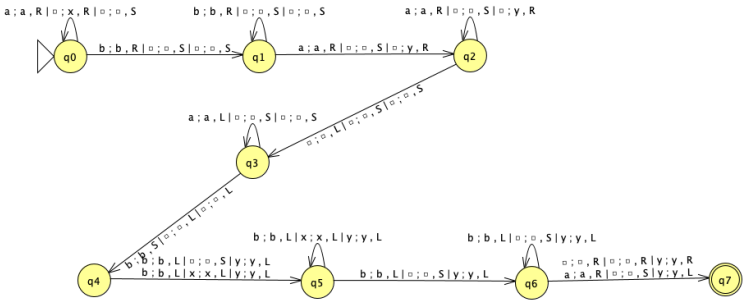
d) $L = \{ a^i b^j a^k \mid i = j$ ou $j = k \}$ fazer com 1 fita e com 3 fitas. Σ = {a,b}



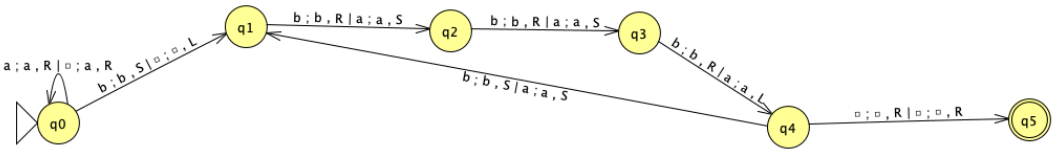


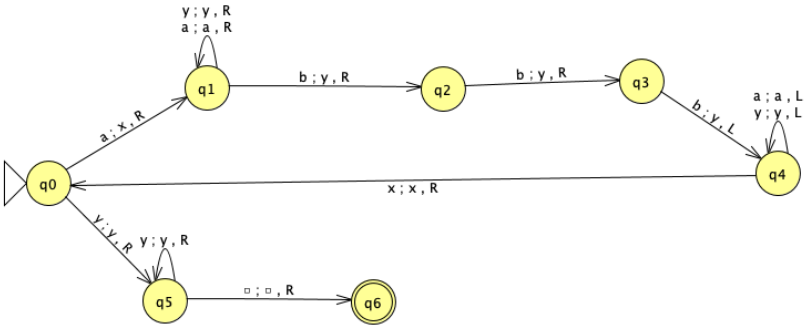
e) $L = \{ a^i b^j a^k \mid i < j < k \}$ fazer com 1 fita e com 3 fitas. Σ = {a,b}





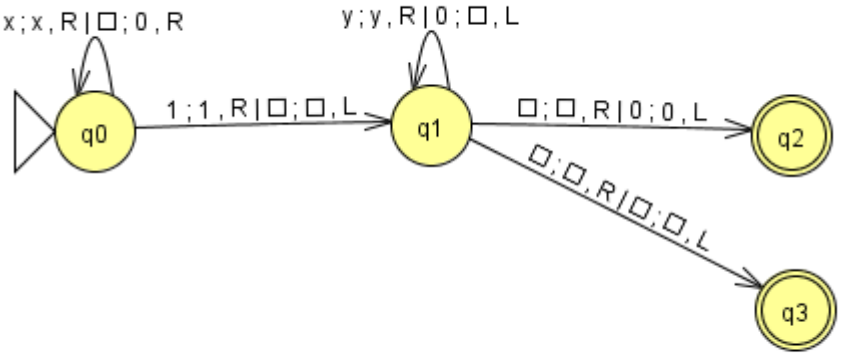
f) $L = \{ a^n b^{3n} : n \geq 1 \}$ fazer com 1 fita e com 2 fitas. Σ = {a,b}



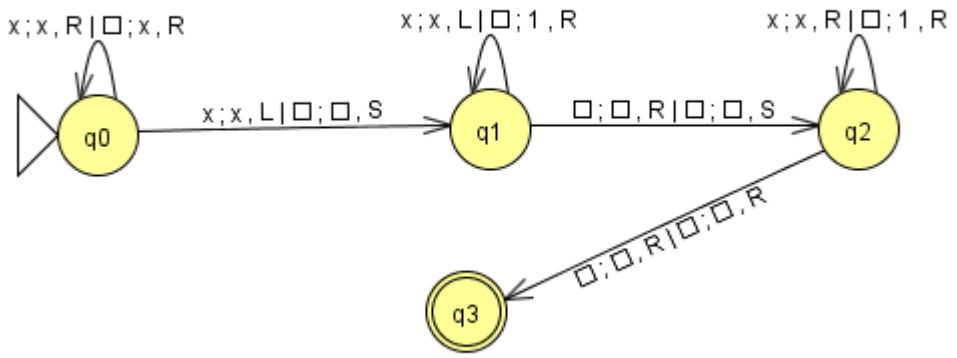


9) Projete Máquinas de Turing que calcule as funções:
Obs: sugestão: usar duas fitas. Avaliar se precisa de mais fitas.

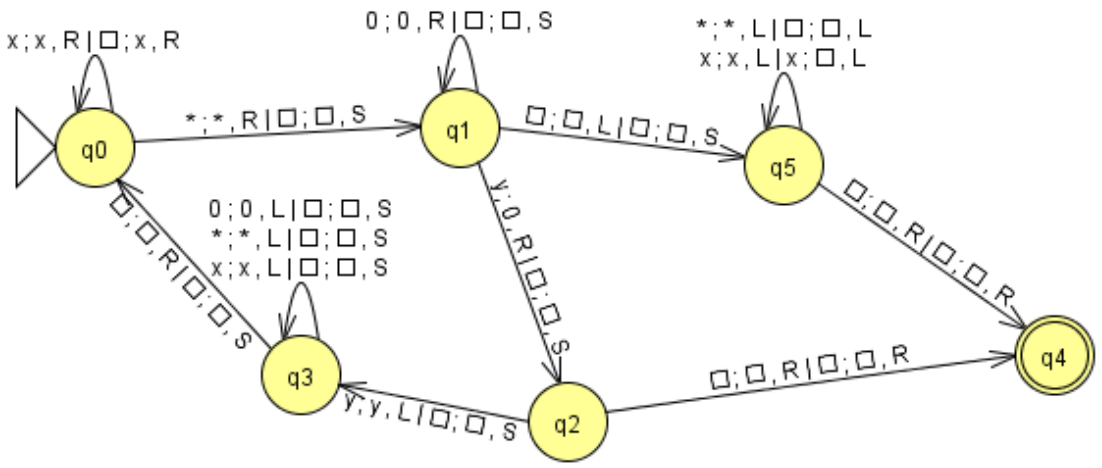
a) f(x,y) = x – y, para x e y inteiros e positivos, com x > y



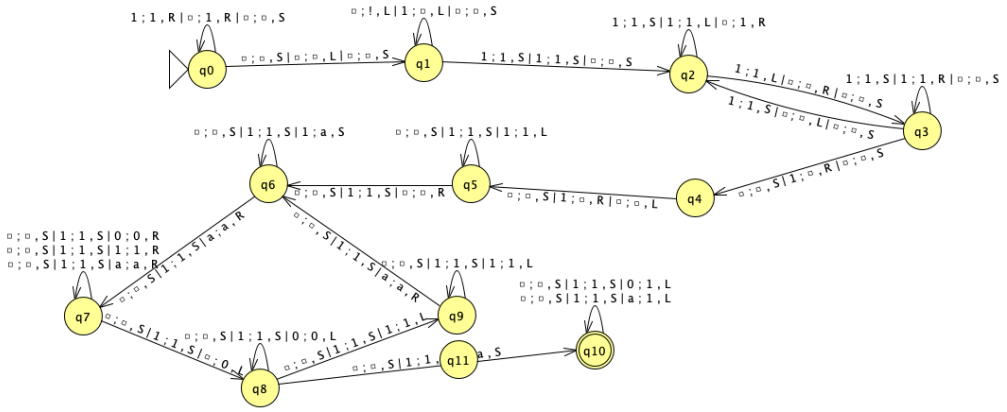
b) f(x,y) = 3*x, para x inteiro e positivo.



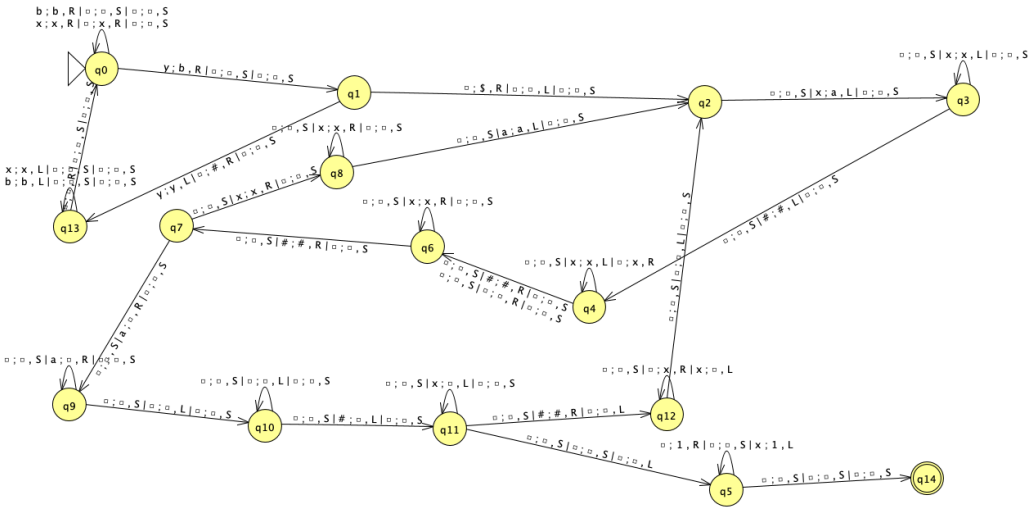
c) f(x,y) = x*y, para x e y inteiros e positivos



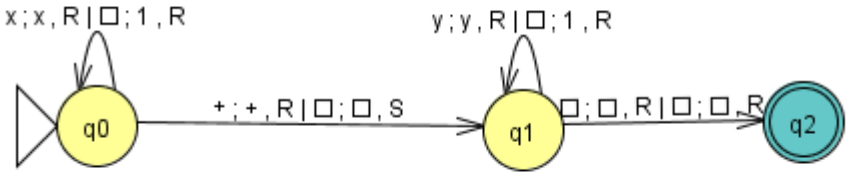
d) fatorial de A, sendo A um número inteiro positivo



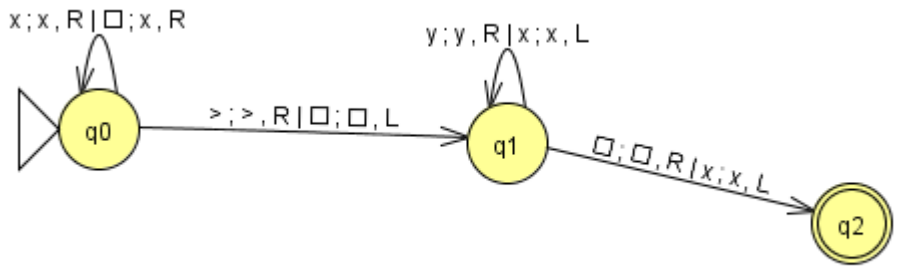
e) pot(x,y) = (x ^y), para x e y inteiros positivos



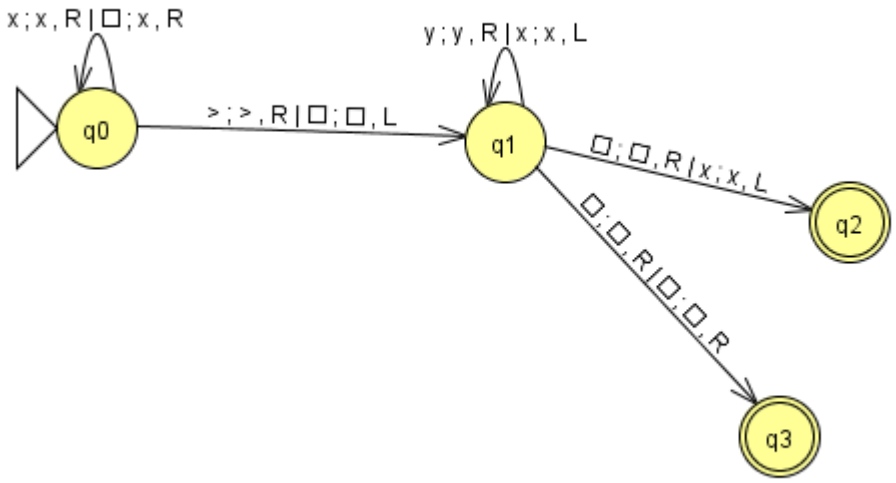
f) f(x,y) = x + y para x e y inteiros positivos



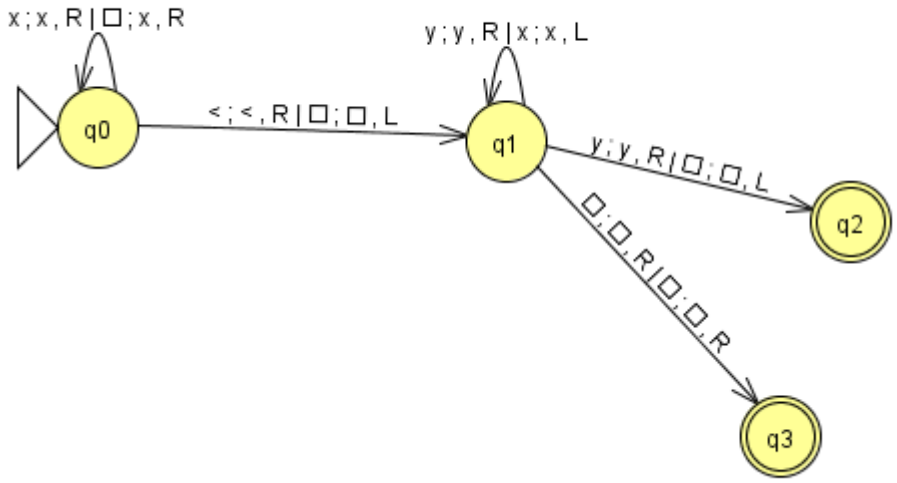
g) teste(A > B): nos inteiros positivos



h) teste (A >= B): nos inteiros positivos



i) teste (A <=B) : nos inteiros positivos

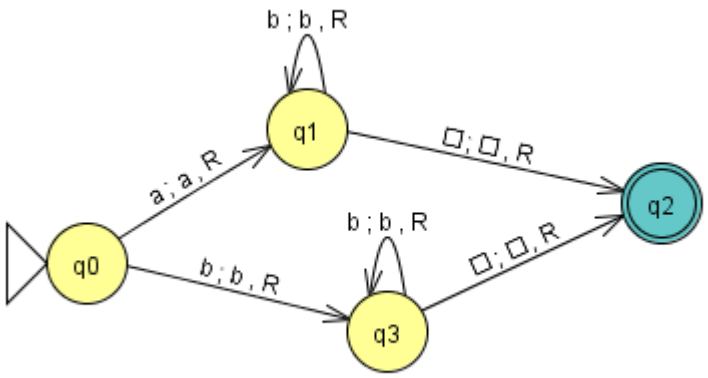


10)

a)

Qual a linguagem aceita pela Máquina de Turing $M = (\{q_0,q_1,q_2,q_3\}, \{a,b\}, \{a,b,\square\}, \delta, q_0, \{q_3\})$, onde:

$\delta(q_0, a) = (q_1, a, R)$,
$\delta(q_0, b) = (q_2, b, R)$,
$\delta(q_1, b) = (q_1, b, R)$,
$\delta(q_1, \square) = (q_3, \square, R)$,
$\delta(q_2, a) = (q_3, a, R)$,
$\delta(q_2, b) = (q_2, b, R)$.

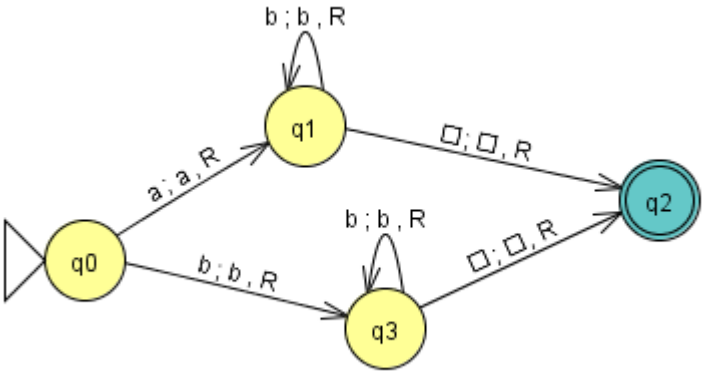


Linguagem: {ab* | bb*a}

b)

$$\Sigma = \{a, b\}, \Gamma = \{a, b, \square\}, F = \{q_3\}, Q = \{q_0, q_1, q_2, q_3\}$$

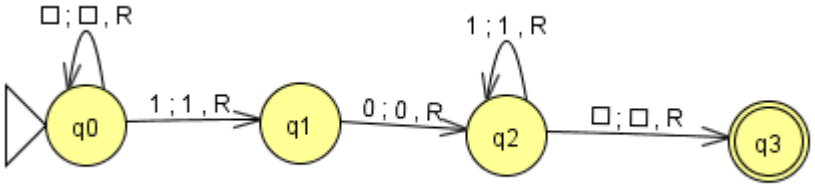
$$\delta(q_0, a) = (q_1, a, R),$$
$$\delta(q_0, b) = (q_2, b, R),$$
$$\delta(q_1, b) = (q_1, b, R),$$
$$\delta(q_1, \square) = (q_3, \square, R),$$
$$\delta(q_2, b) = (q_2, b, R),$$
$$\delta(q_2, a) = (q_3, a, R).$$



Linguagem: {ab* | bb*a}

c)

$$\Sigma = \{0,1\}, \Gamma = \{0,1,\square\}, F = \{q_f\}, Q = \{q_0, q_1, q_2, q_f\}$$
$$\delta(q_0, \square) = (q_0, \square, R),\ \delta(q_0, 1) = (q_1, 1, R),$$
$$\delta(q_1, 0) = (q_2, 0, R),$$
$$\delta(q_2, 1) = (q_2, 1, R),$$
$$\delta(q_2, \square) = (q_f, \square, R).$$

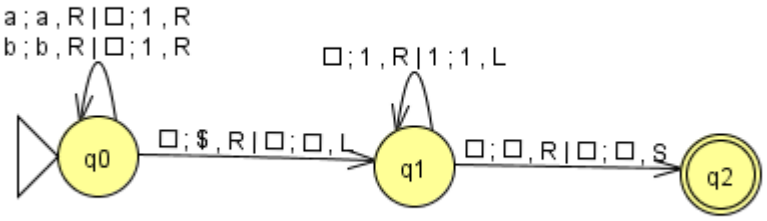


Linguagem: {101*}

11)

Construa uma Máquina de Turing que calcule o tamanho de uma cadeia w, onde w ∈ {a,b}*, onde o tamanho deverá estar escrito na base unária. (A Máquina de Turing simulará o comando **strlen()** da linguagem C). Por exemplo: entrada = abaaba, saída = abaaba$111111.

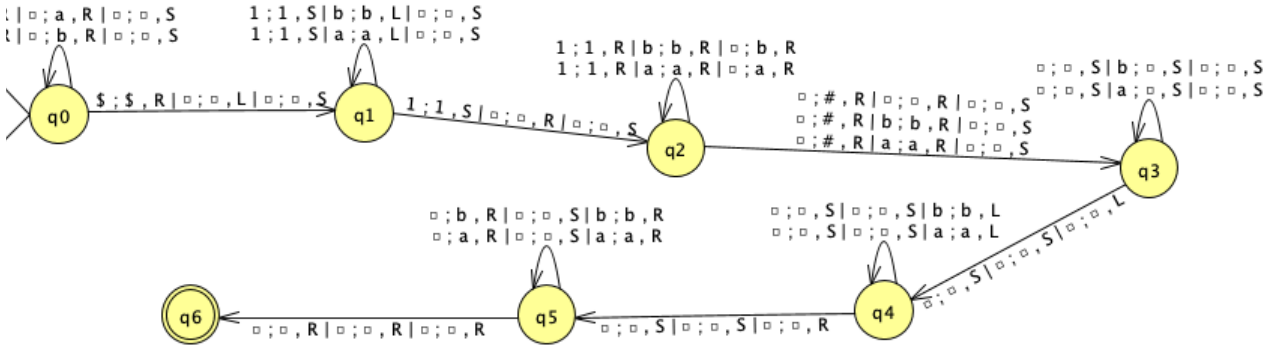
Usar quantas fitas achar necessário.



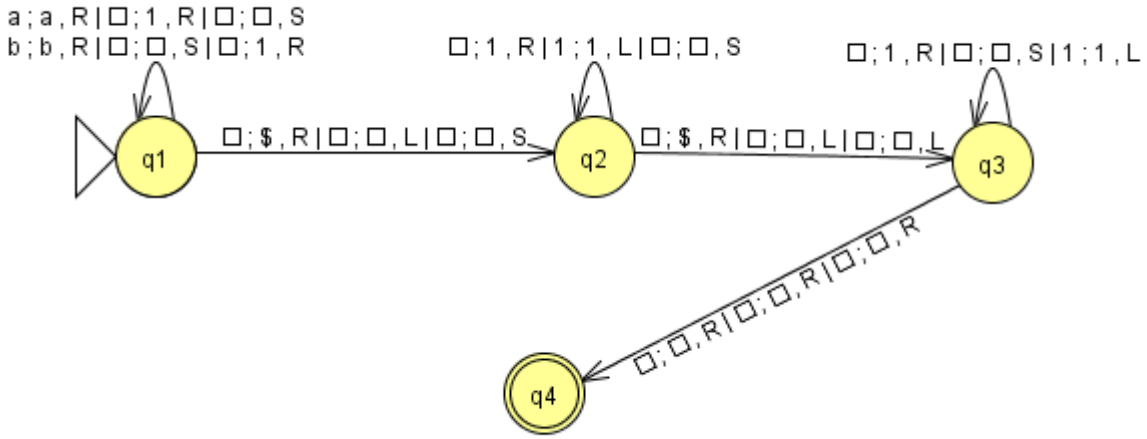
12)

Construa uma Máquina de Turing que simule o comando **strncpy(w1, w2, qt)** da linguagem C, sabendo que as cadeias são do tipo {a,b}* e a quantidade estará escrita em base unária. Por exemplo, se entrada = aaabbbaa$1111, então a saída = aaabbbaa$1111#aaab

Usar quantas fitas achar necessário.



13)

Construa uma Máquina de Turing que trabalha sobre cadeias do tipo {a,b}* e conte quantas letras a's e b's aparecem nesta cadeia. O número deverá estar na base unária. Por exemplo, se entrada = aaabbbaa, então a saída = aaabbbaabb$11111$111.

Usar quantas fitas achar necessário.